

STEPHENVILLE, NL, Canada

WMO: 718150

Lat: 48.5614N

Lon: 58.5661W

Elev: 25

StdP: 101.03

Time Zone: -3.50 (NAF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.8	-13.7	-19.9	0.6	-15.1	-17.3	0.8	-13.1	18.5	-4.9	16.4	-5.5	5.1	280	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	23.4	18.3	22.1	17.7	20.9	17.3	19.6	21.9	18.8	21.0	18.1	20.1	4.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.5	21.0	17.8	12.8	20.3	17.1	12.3	19.6	56.1	22.0	53.6	21.1	51.3	20.2	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.9	11.8	10.1	DB	-19.0	27.0	2.7	1.5	-20.9	28.0	-22.5	28.9	-24.0	29.7	-25.9	30.8	(4)
(5)			WB	-19.4	21.4	2.7	1.2	-21.3	22.3	-22.9	22.9	-24.4	23.6	-26.3	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-5.1	-5.8	-3.1	2.2	7.5	12.1	16.6	17.1	13.0	7.8	3.0	-1.6	(6)
(7)		DBStd	8.68	4.29	4.81	4.63	3.59	3.27	3.25	2.43	2.20	3.18	3.29	3.80	4.03	(7)
(8)		HDD10.0	2325	469	441	405	236	93	14	0	0	9	86	211	361	(8)
(9)		HDD18.3	4756	728	674	663	486	337	188	64	49	162	327	459	619	(9)
(10)		CDD10.0	636	0	0	0	1	14	78	204	221	99	17	2	0	(10)
(11)		CDD18.3	25	0	0	0	0	0	1	11	12	2	0	0	0	(11)
(12)		CDH23.3	45	0	0	0	0	0	6	21	16	2	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	4.9	6.2	5.9	5.4	5.1	4.4	3.8	3.6	3.9	4.3	4.9	5.3	5.9	(14)
(15)	Precipitation	PrecAvg	1364	134	107	96	74	103	104	113	121	125	129	125	128	(15)
(16)		PrecMax	1784	268	260	161	142	253	223	219	352	192	219	207	245	(16)
(17)		PrecMin	1169	46	40	39	28	48	28	43	30	73	50	71	43	(17)
(18)		PrecStd	147	48	45	31	31	45	46	47	57	34	45	32	41	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	8.8	8.5	14.1	19.3	23.9	25.6	25.1	22.8	18.9	15.3	10.4	(19)
(20)			MCWB	7.5	7.5	6.2	10.1	12.3	16.3	19.1	18.9	18.5	15.9	13.5	9.1	(20)
(21)		2%	DB	5.3	5.5	6.4	11.5	16.7	21.0	23.4	23.3	20.6	16.2	13.0	7.8	(21)
(22)			MCWB	4.4	4.2	4.6	8.1	11.8	15.7	18.3	18.7	17.5	13.7	11.5	6.8	(22)
(23)		5%	DB	3.0	2.9	4.7	9.3	14.9	19.2	22.1	22.1	19.2	14.6	10.9	5.7	(23)
(24)			MCWB	2.1	1.6	3.0	6.5	11.1	14.9	17.5	18.1	16.6	12.4	9.7	4.6	(24)
(25)		10%	DB	1.1	1.0	3.0	7.5	13.2	17.6	20.7	21.1	18.1	13.0	8.7	3.9	(25)
(26)			MCWB	0.2	0.0	1.4	5.0	9.8	14.0	17.2	17.7	15.7	11.1	7.5	2.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	7.5	6.7	11.1	14.7	18.6	20.9	20.7	19.6	16.4	13.8	9.4	(27)
(28)			MCD B	8.4	8.6	7.9	12.8	17.0	21.4	24.3	22.9	22.0	18.1	15.2	10.3	(28)
(29)		2%	WB	4.5	4.4	4.9	8.8	13.0	17.1	19.4	19.7	18.2	14.2	11.6	6.8	(29)
(30)			MCD B	5.3	5.2	6.3	10.9	15.6	19.5	21.9	21.9	20.0	15.8	12.8	7.7	(30)
(31)		5%	WB	2.2	1.7	3.2	6.9	11.5	15.7	18.6	19.0	17.2	12.7	9.8	4.7	(31)
(32)			MCD B	3.0	2.7	4.5	8.7	14.3	18.3	20.8	21.2	18.8	14.3	10.9	5.6	(32)
(33)		10%	WB	0.3	0.0	1.5	5.2	10.0	14.3	17.8	18.3	16.0	11.3	7.6	3.0	(33)
(34)			MCD B	1.0	0.9	2.8	7.2	12.9	17.0	19.9	20.5	17.7	12.8	8.6	3.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	6.6	7.0	6.9	8.2	8.1	7.1	7.2	7.2	6.4	5.5	5.3	(35)
(36)			MCD BR	8.8	8.4	8.1	9.5	11.1	10.7	9.4	8.5	8.0	8.4	8.3	7.9	(36)
(37)			MCWBR	8.5	7.7	6.9	6.9	7.2	6.4	5.5	5.3	5.9	6.5	7.8	7.6	(37)
(38)		5% WB	MCD BR	9.0	8.4	8.0	9.0	9.5	9.1	7.0	7.1	7.5	8.3	7.9	(38)	
(39)			MCWBR	8.8	8.0	7.3	7.4	7.0	6.3	4.8	5.1	5.9	6.5	7.9	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.301	0.326	0.351	0.365	0.376	0.411	0.396	0.370	0.343	0.316	0.298	(40)	
(41)		taud	2.314	2.359	2.351	2.387	2.422	2.417	2.359	2.400	2.453	2.478	2.468	2.378	(41)	
(42)		Ebn at Noon	751	855	889	899	897	885	846	844	833	799	744	713	(42)	
(43)		Edh at Noon	72	86	102	110	111	112	117	108	93	77	63	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.70	1.33	2.74	3.93	5.10	5.38	5.14	4.54	3.35	2.06	1.00	0.56	(44)	
(45)		RadStd	0.08	0.14	0.29	0.40	0.35	0.51	0.43	0.24	0.26	0.14	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	+0.33	+0.37	N/A	N/A	N/A	N/A	(46)
(47) Regional (9 neighbors)		N/A	N/A	+1.27	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page